

Project Report

Wildlife Information System (WIS)

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4. DESIGN

4.1 Class Diagram

The class diagram represents the main classes in the system with their attributes and methods:

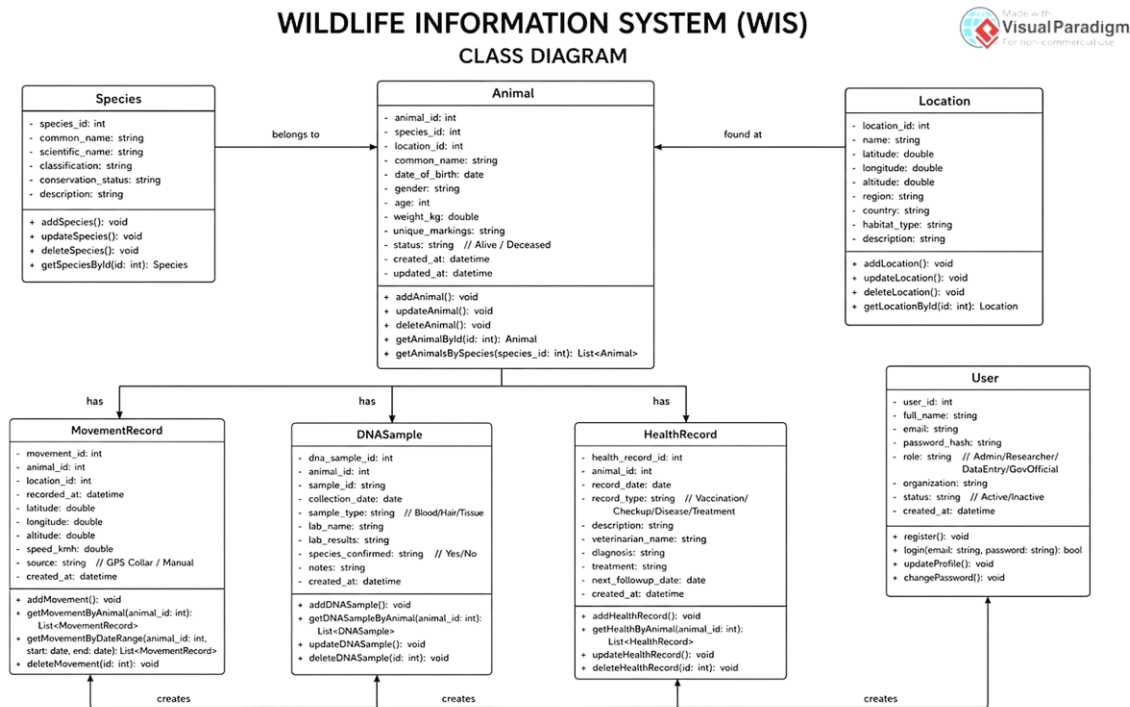


Figure 9: Class Diagram

Class: Animal

Attributes: animalId, speciesId, name, gender, ageYears, weightKg, locationId, addedBy, addedAt

Methods: save(), update(), delete(), getMovementHistory(), getHealthRecords(), getDNASamples()

Class: MovementRecord

Attributes: recordId, animalId, latitude, longitude, recordedAt, source

Methods: save(), getByAnimal(), bulkInsert()

Class: DNASample

Attributes: sampleId, animalId, collectedAt, labName, result, speciesConfirmed

Methods: save(), getByAnimal()

Class: HealthRecord

Attributes: healthId, animalId, date, condition, treatment, vetName, vaccineGiven

Methods: save(), update(), getByAnimal()

Class: User

Attributes: userId, name, email, passwordHash, role, createdAt

Methods: login(), logout(), updateProfile(), changePassword()

Class: Species

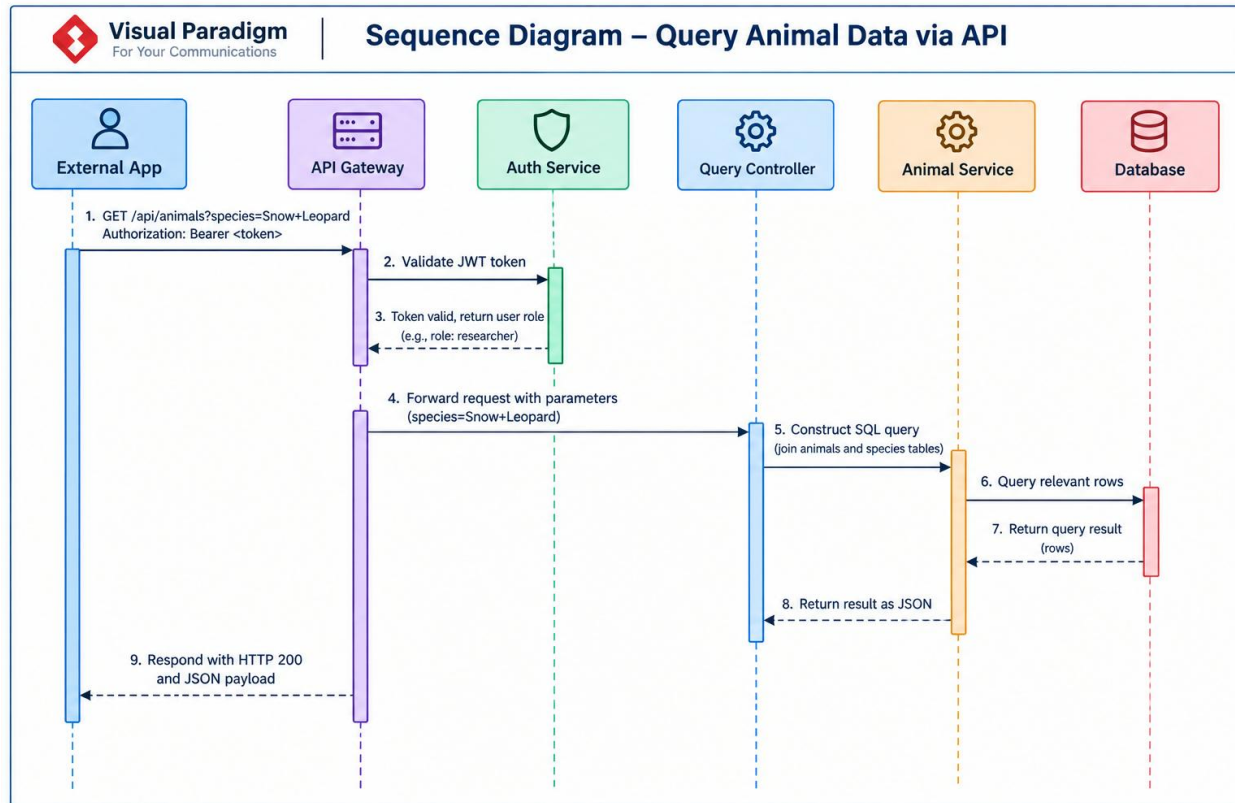
Attributes: speciesId, commonName, scientificName, conservationStatus, habitat

Methods: save(), getAll(), getName()

4.2 Sequence Diagram

The detailed sequence for "Query Animal Data via API" is described below:

1. External App sends: GET /api/animals?species=Snow+Leopard with Authorization: Bearer <token>
2. The API Gateway validates the JWT token.
3. The Auth Service validates the token and provides the user role.
4. The Query Controller handles the incoming request and forwards the parameters to the Animal Service.
5. The Animal Service constructs an SQL query that queries both the animals table and the species table.
6. The database queries the relevant rows.
7. The Animal Service returns the result as JSON.
8. The API Gateway responds with an HTTP 200 and the JSON payload.



4.3 Architecture

The Wildlife Information System follows a 3-Tier Architecture:

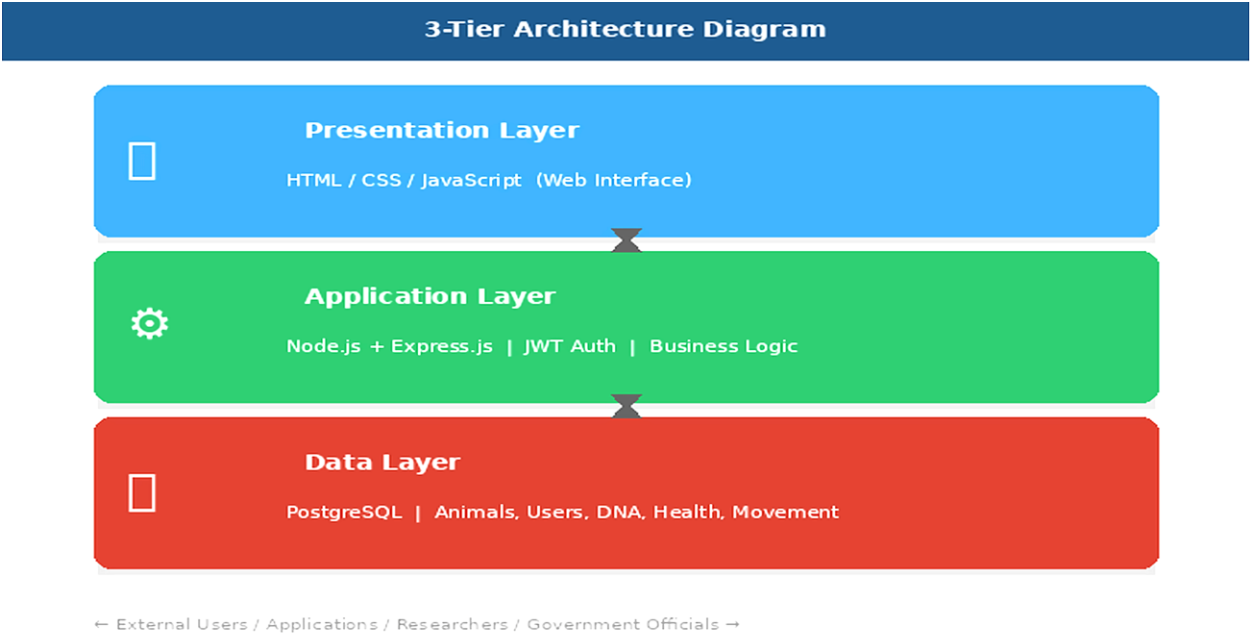


Figure 10: 3-Tier Architecture Diagram

Layer	Component	Responsibility
Presentation Layer	HTML/CSS/JS Web Interface	Basic UI for data entry, search, and export
Application Layer	Node.js + Express.js API Server	Business logic, authentication, query processing
Data Layer	PostgreSQL Database	Stores all wildlife records, users, and audit logs

External users and applications communicate with the Application Layer through the RESTful API. The Application Layer handles authentication, business logic, and data formatting. The Data Layer stores everything persistently

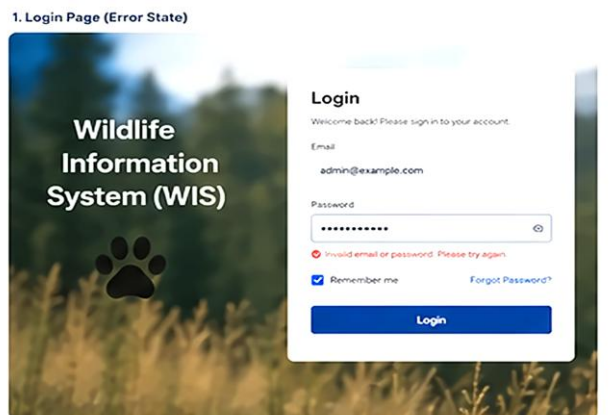
Key API Endpoints

Method	Endpoint	Description
GET	/api/animals	Get list of all animals (with filters)
GET	/api/animals/:id	Get a single animal by ID
POST	/api/animals	Add a new animal record
PUT	/api/animals/:id	Update animal record
DELETE	/api/animals/:id	Delete animal record
GET	/api/animals/:id/movements	Get movement history for an animal
POST	/api/movements/upload	Bulk upload CSV movement data
GET	/api/animals/:id/health	Get health records for an animal
GET	/api/animals/:id/dna	Get DNA records for an animal

Method	Endpoint	Description
POST	/api/auth/login	User login and get JWT token
GET	/api/species	Get list of all species

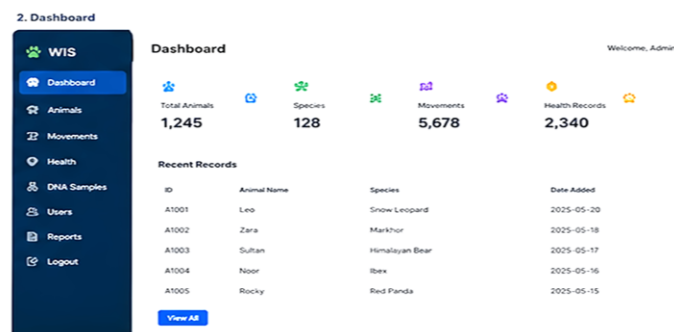
4.4 Interface Design

This application would have an easy-to-use web-based interface. The key pages include:



Login Page : Contains the email address field, password field, and login button. Displays an error message if incorrect.

Dashboard : Has a welcome message, statistics about animals (including total number, number of species, and some recent records), and navigation menu.



Animal List Page : Displays a sortable table containing all animals with View, Edit, and Delete buttons.

3. Animal List Page

ID	Name	Species	Gender	Age	Location	Actions
A1001	Leo	Snow Leopard	M	5	Gilgit	[Edit] [Delete] [Details]
A1002	Zera	Markhor	F	3	Chitral	[Edit] [Delete] [Details]
A1003	Sultan	Himalayan Bear	M	6	Skardu	[Edit] [Delete] [Details]
A1004	Nisar	Ibex	F	4	Muzee	[Edit] [Delete] [Details]
A1005	Rocky	Red Panda	M	2	Islamabad	[Edit] [Delete] [Details]
A1006	Beta	Brown Bear	F	7	Hunza	[Edit] [Delete] [Details]
A1007	Shadow	Wolf	M	4	Astore	[Edit] [Delete] [Details]
A1008	Simba	Lion	M	6	Bahawalpur	[Edit] [Delete] [Details]

Showing 1 to 8 of 255 animals

Add Animal Form : Contains fields for entering information on the species, name, gender, age, weight, and location of an animal. Includes a submission button for recording the data.

4. Add Animal Form

Add New Animal

Species: Select Species

Name: Enter animal name

Gender: Select Gender

Age (Years): Enter age

Weight (kg): Enter weight

Location: Select Location

Movement Records Page: Displays a sortable table containing GPS coordinates and the time stamps for a particular animal with an option to add a CSV file.

5. Movement Records Page

Record ID	Latitude	Longitude	Recorded At
M5001	35.1234	74.5678	2025-05-20 08:10:30
M5002	35.2234	74.6678	2025-05-20 12:45:20
M5003	35.3234	74.7678	2025-05-20 16:10:50
M5004	35.4234	74.8678	2025-05-20 20:30:15
M5005	35.5234	74.9678	2025-05-21 00:50:05

Showing 1 to 5 of 320 records

API Documentation Page: A straightforward page listing APIs, required parameters, and example responses.

User Management (Admin Only): Displays a sortable table containing information on all the users registered with options to create or remove an account.

Design principles: Minimal use of color; the only colors being used are blue and white. All buttons have descriptive names. Errors are displayed in red. Success messages are displayed in green.

6. API Documentation Page

API Documentation

Base URL
https://api.wis.com/v1

Endpoints

Method	Endpoint	Description
GET	/animals	Get list of all animals
GET	/animals/{id}	Get details of a specific animal
POST	/animals	Create a new animal
PUT	/animals/{id}	Update an existing animal
DELETE	/animals/{id}	Delete an animal

Example: Get Animal Details

GET /animals/{id}

Parameters

Name	Type	Description
id	integer	Animal ID

Response (200 OK)

```
{  "id": "A1001",  "name": "Leo",  "species": "Snow Leopard",  "gender": "M",  "age": 5,  "weight": 45,  "location": "Hilltop"}
```

7. User Management (Admin Only)

WIS

Dashboard

Animals

Movements

Health

DNA Samples

Users

Reports

Logout

Users

+ Create User

ID	Full Name	Email	Role	Status	Created At	Actions
U001	Admin User	admin@example.com	Administrator	Active	2025-01-10 09:30:00	
U002	Ranger One	ranger1@example.com	Ranger	Active	2025-01-12 10:15:20	
U003	Vet Doctor	vet@example.com	Veterinarian	Active	2025-01-14 11:45:10	
U004	Researcher	research@example.com	Researcher	Inactive	2025-01-18 14:20:30	
U005	Viewer	viewer@example.com	Viewer	Active	2025-01-20 16:05:40	

Showing 1 to 5 of 25 users

1

2

3

4

5

...

Create User

Full Name

Enter full name

Email

Enter email

Role

Select Role

Password

Enter password

Confirm Password

Confirm password

Status

Active

Create User

Cancel

Edit User (U002)

Full Name

Ranger One

Email

ranger1@example.com

Role

Ranger

Status

Active

Update User

Cancel